① srcML of lines 14 and 18–19

```
14 qc.cx(q[src], q[a])
```

Python

```
<expr_stmt pos:start="14:5"><expr><call><name><name>qc</name><operator>.</operator><name>cx</name></name>
    ↪<argument_list>(<argument><expr><name><name>q</name><index>[<expr><name>src</name></expr>]</index></name>
    ↪</expr></argument>, <argument><expr><name><name>q</name><index>[<expr><name>a</name></expr>]</index></name>
    ↪</expr></argument>)</argument_list></call></expr></expr_stmt>
```

srcML

```
18 with qc.if_test((c[2 * h + 1], 1)):
19     qc.x(q[b])
```

Python

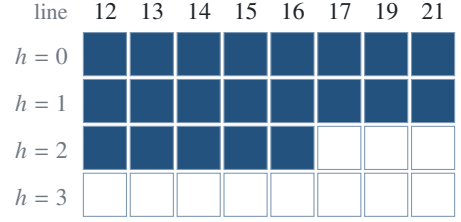
```
<with pos:start="18:5">with <expr><call><name><name>qc</name><operator>.</operator><name>if_test</name>
    ↪</name><argument_list>(<argument><expr><tuple>(<expr><name><name>c</name><index>[<expr>
    ↪<literal type="number">2</literal><operator>*</operator><name>h</name><operator>+</operator>
    ↪<literal type="number">1</literal></expr>]</index></name></expr>, <expr><literal type="number">1</literal>
    ↪</expr>)</tuple></expr></argument>)</argument_list></call></expr><block>:<block_content>
        <expr_stmt><expr><call><name><name>qc</name><operator>.</operator><name>x</name></name>
            ↪<argument_list>(<argument><expr><name><name>q</name><index>[<expr><name>b</name></expr>]</index>
            ↪</name></expr></argument>)</argument_list></call></expr></expr_stmt>
</block_content></block></with>
```

srcML

`cx` gate `q[src]` qubit argument, evaluated in each iteration `c` classical bit, written by a measurement and read by a condition `if_test` conditioned block `pos:start` source line

② Trace of line 8 and iteration $h = 0$ ⟶ ③ Slice of `q[4]` by line and iteration

line	h	operation
8	–	u q_0
12	0	h q_1
13	0	cx q_1, q_2
14	0	cx q_0, q_1
15	0	h q_0
16	0	measure $q_0 \to c_0$
17	0	measure $q_1 \to c_1$
19	0	x q_2 if c_1
21	0	z q_2 if c_0
	1–3	the same on the next qubits



line 8 in the slice, line 23 not **22** of 34 operations